# THE COH CATEGORY V3: A LAWVERE QUASI-METRIC INDUCED BY CERTIFIED OPLAX STRUCTURE

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Abstract. We define Coh Category V3 as a metric-enriched extension of the projection-certified Coh V2 category. The metric is not postulated; it is induced from admissible traces by taking the infimum of a projection-derived semantic defect envelope. Starting from a hidden realization layer, an observable trace layer, a projection map, and a hidden cost functional, V2 constructs certified morphisms whose declared defect dominates the worst-case hidden cost over each observable fiber. V3 adds an admissible path space and defines

$$d_{\mathrm{Coh}}(x, y) = \inf_{\tau \in \mathcal{T}(x,y)} \delta(\tau),$$

with value $+\infty$ when no admissible trace exists. We prove that this defines a directed extended pseudo-quasi-metric and, equivalently, a category enriched over the Lawvere quantale $([0, \infty], \geq, +, 0)$. The result is a geometry of certified computation: distance is the minimal projection-induced defect required to pass from one state to another through verified morphisms. We also isolate the exact assumptions required for the construction, record failure modes, and describe the geodesic and growth structures naturally induced by V3.

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## 1. Introduction

The first two layers of the Coh program separate execution, certification, and projection. V1 is the operational verifier layer: a proposed transition becomes admissible only after satisfying a deterministic acceptance rule and a resource inequality. V2 lifts this to a projection-aware certified category: hidden realizations project to observable traces, and an envelope defect records the worst-case hidden cost over each observable fiber. V3 should not redefine either of those layers. Its role is to add geometry on top of the V2 category.

The governing idea is the following.

$$\boxed{\text{V3 turns certified morphisms into a directed admissible geometry.}}$$

The distance from $x$ to $y$ is not externally chosen. It is the smallest projection-induced defect needed to travel from $x$ to $y$ through admissible traces. This gives a Lawvere-style directed distance rather than a symmetric metric. Directionality is essential: resource inequalities, verifier acceptance, and hidden realization costs generally distinguish $x \to y$ from $y \to x$.

**Contributions.** This paper makes the following precise claims.

(1) It defines the explicit admissible trace domain $\mathcal{T}(x, y)$ and handles the empty case by assigning distance $+\infty$.
(2) It proves identity, nonnegativity, and the directed triangle inequality for $d_{\mathrm{Coh}}$.
(3) It classifies the V3 structure as a Lawvere quasi-metric category, equivalently an enrichment over $([0, \infty], \geq, +, 0)$.
(4) It isolates the assumptions required for the result: V2 closure, identity tightness, and an oplax defect law.
(5) It defines geodesics, metric balls, and growth as derived V3 structures, not as additional axioms.

**Status of claims.** All categorical and metric statements in the body are proved from explicit axioms. Statements about geodesic existence require compactness or finiteness assumptions and are therefore marked as conditional. Growth dimension is a derived observable of the induced geometry, not a universal theorem.

## 2. Version stack

The version stack is summarized in Table 1.

| Layer | Role | Mathematical output |
|---|---|---|
| V1 | operational admissibility | accepted traces and telescoping resource law |
| V2 | projection-certified category | certified morphisms and oplax envelope defect |
| V3 | induced geometry | Lawvere quasi-metric and admissible geometry |
| V4 | dynamics on V3 | PDE/control coupling and physical evolution |

Table 1. Coh version stack. V3 is not a replacement for V2; it is the metric enrichment induced by V2.

## 3. V2 substrate recalled

We begin with the V2 projection-certified substrate. The definitions in this section are included only to fix notation and dependencies for V3.

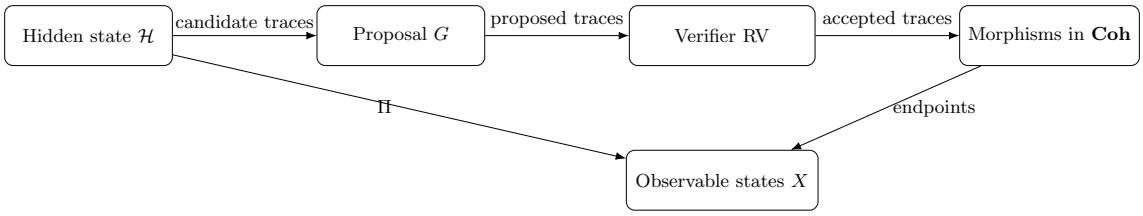



FIGURE 1. Coh execution wedge. Candidate traces are proposed, certified by a deterministic verifier, and only certified traces enter the categorical layer. V3 measures distances inside this certified layer.

**Definition 3.1** (Hidden substrate)**.** *A hidden substrate consists of a class of hidden realizations $\mathcal{G}$ together with a partially defined concatenation operation*

$$\cdot : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}.$$

*When $\xi_2 \cdot \xi_1$ is defined, $\xi_1$ is executed first and $\xi_2$ second.*

**Definition 3.2** (Observable trace substrate)**.** *An observable trace substrate consists of a class $\mathcal{V}$ of observable traces, a partially defined composition operation*

$$\odot : \mathcal{V} \times \mathcal{V} \rightharpoonup \mathcal{V},$$

*and an identity trace $e \in \mathcal{V}$.*

**Definition 3.3** (Projection-cost layer)**.** *A projection-cost layer consists of a projection map*

$$\Pi : \mathcal{G} \to \mathcal{V}$$

*and a hidden cost functional*

$$W : \mathcal{G} \to [0, \infty].$$

*The extended codomain is allowed so that unbounded fibers can be recorded honestly rather than hidden.*

**Axiom 3.4** (Observable associativity)**.** *For all composable $R_1, R_2, R_3 \in \mathcal{V}$,*

$$(R_3 \odot R_2) \odot R_1 = R_3 \odot (R_2 \odot R_1).$$

**Axiom 3.5** (Observable identity)**.** *For every $R \in \mathcal{V}$,*

$$e \odot R = R, \qquad R \odot e = R$$

*whenever either side is defined.*

**Axiom 3.6** (Projection compatibility)**.** *If $\xi_2 \cdot \xi_1$ is defined, then*

$$\Pi(\xi_2 \cdot \xi_1) = \Pi(\xi_2) \odot \Pi(\xi_1).$$

**Axiom 3.7** (Fiber decomposition)**.** *If $\xi \in \mathcal{G}$ satisfies*

$$\Pi(\xi) = R_2 \odot R_1,$$

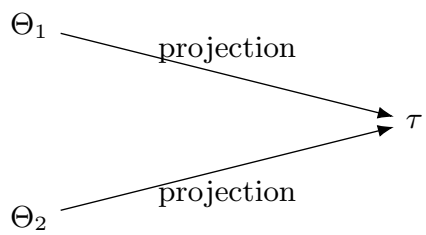
*then there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that*

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

**Axiom 3.8** (Hidden cost subadditivity)**.** *If $\xi_2 \cdot \xi_1$ is defined, then*

$$W(\xi_2 \cdot \xi_1) \leq W(\xi_1) + W(\xi_2).$$

*If equality holds, the hidden layer is additive. V3 only needs the subadditive form.*




multiple hidden realizations can collapse to one observable trace

FIGURE 2. Projection collapse. Multiple hidden realizations can project to a single observable trace. V2 and V3 therefore reason by the worst-case fiber envelope $\delta(R) = \sup_{\xi \in \mathcal{F}_R} W(\xi)$.

## 4. Envelope defect

**Definition 4.1** (Observable fiber)**.** *For $R \in \mathcal{V}$, define*

$$\mathcal{F}_R := \{\xi \in \mathcal{G} : \Pi(\xi) = R\}.$$

**Definition 4.2** (Admissible observable trace)**.** *An observable trace $R \in \mathcal{V}$ is fiber-admissible if $\mathcal{F}_R \neq \emptyset$. Write $\mathcal{V}_{\mathrm{adm}}$ for the class of fiber-admissible traces.*

**Definition 4.3** (Envelope defect)**.** *For $R \in \mathcal{V}_{\mathrm{adm}}$, define*

$$\delta(R) := \sup_{\xi \in \mathcal{F}_R} W(\xi) \in [0, \infty].$$

*If $R \notin \mathcal{V}_{\mathrm{adm}}$, $\delta(R)$ is left undefined unless explicitly extended by $+\infty$.*

**Assumption 4.4** (Finite envelope option)**.** *When finite certification is required, assume*

$$\sup_{\xi \in \mathcal{F}_R} W(\xi) < \infty$$

*for every trace $R$ admitted into the certified category. Otherwise all statements are interpreted in the extended domain $[0, \infty]$.*

**Lemma 4.5** (Envelope soundness)**.** *If $\xi \in \mathcal{G}$ and $\Pi(\xi) \in \mathcal{V}_{\mathrm{adm}}$, then*

$$W(\xi) \leq \delta(\Pi(\xi)).$$

*Proof.* By definition, $\xi \in \mathcal{F}_{\Pi(\xi)}$. The value $\delta(\Pi(\xi))$ is the supremum of $W$ over this fiber, so it is an upper bound for $W(\xi)$. □

**Lemma 4.6** (Envelope minimality)**.** *Let $\Delta : \mathcal{V}_{\mathrm{adm}} \to [0, \infty]$ satisfy*

$$W(\xi) \leq \Delta(\Pi(\xi))$$

*for every hidden realization whose projection is admissible. Then for every $R \in \mathcal{V}_{\mathrm{adm}}$,*

$$\delta(R) \leq \Delta(R).$$

*Proof.* Fix $R$. For every $\xi \in \mathcal{F}_R$, one has $\Pi(\xi) = R$, and therefore $W(\xi) \leq \Delta(R)$. Hence $\Delta(R)$ is an upper bound of $\{W(\xi) : \xi \in \mathcal{F}_R\}$. Taking the supremum gives $\delta(R) \leq \Delta(R)$. □

**Theorem 4.7** (Projection-induced oplax defect)**.** *Let $R_1, R_2 \in \mathcal{V}_{\mathrm{adm}}$, and assume $R_2 \odot R_1$ is admissible. Then*

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2).$$

$$\begin{tikzcd} x \arrow[r, "\tau_1"] \arrow[rr, bend left, "\tau_2 \circ \tau_1 = \tau_2 * \tau_1"] & y \arrow[r, "\tau_2"] & z \end{tikzcd}$$

FIGURE 3. Certified composition. The categorical composite is trace concatenation. The V3 distance is obtained by minimizing the envelope defect over such admissible traces.

*Proof.* Let $\xi \in \mathcal{F}_{R_2 \odot R_1}$. By fiber decomposition, there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

Thus $\xi_i \in \mathcal{F}_{R_i}$. Hidden cost subadditivity gives

$$W(\xi) = W(\xi_2 \cdot \xi_1) \leq W(\xi_1) + W(\xi_2) \leq \delta(R_1) + \delta(R_2).$$

The right-hand side is an upper bound for $W$ over $\mathcal{F}_{R_2 \odot R_1}$. Taking the supremum over this fiber yields the result. □

## 5. Certified morphisms inherited from V2

**Definition 5.1** (Boundary object)**.** *A boundary object $X$ consists of a carrier set $\Sigma_X$ and a valuation*

$$V_X : \Sigma_X \to [0, \infty].$$

*In finite certification contexts, restrict to finite valuation values.*

**Definition 5.2** (Endpoint typing)**.** *For $x \in \Sigma_X$ and $y \in \Sigma_Y$, a trace $\tau \in \mathcal{V}_{\mathrm{adm}}$ may be typed*

$$\tau : x \rightsquigarrow y.$$

*This typing relation is primitive at the V2/V3 level.*

**Definition 5.3** (Certified morphism)**.** *A certified morphism from $x$ to $y$ is a tuple*

$$f = (\tau_f, s_f, d_f, x, y)$$

*such that*

*(1) $\tau_f : x \rightsquigarrow y$ is an admissible observable trace;*
*(2) $d_f \geq \delta(\tau_f)$;*
*(3) the Coh resource inequality holds:*

$$V_Y(y) + s_f \leq V_X(x) + d_f.$$

*Here $s_f$ is declared spend and $d_f$ is declared defect.*

**Definition 5.4** (Composition)**.** *For certified morphisms*

$$f = (\tau_f, s_f, d_f, x, y), \qquad g = (\tau_g, s_g, d_g, y, z),$$

*define*

$$g \circ f := (\tau_g \odot \tau_f,\ s_f + s_g,\ d_f + d_g,\ x, z)$$

*whenever the composite trace is defined, admissible, and typed $\tau_g \odot \tau_f : x \rightsquigarrow z$.*

**Proposition 5.5** (V2 category inheritance)**.** *Under V2 closure, certified morphisms form a category. In particular, identities and composites remain certified.*

*Proof.* This is the V2 category theorem. Closure follows because the composite defect declaration satisfies

$$d_f + d_g \geq \delta(\tau_f) + \delta(\tau_g) \geq \delta(\tau_g \odot \tau_f)$$

by the oplax defect law. The resource inequality composes by adding the two certified inequalities and cancelling the finite intermediate valuation. Associativity follows from associativity of observable trace composition, and unit laws follow from the identity trace. □

## 6. V3 admissible trace domain

The following definition is the first genuinely V3 object.

**Definition 6.1** (Admissible trace domain)**.** *For boundary states $x, y$, define*

$$\mathcal{T}(x, y) := \{\tau : x \rightsquigarrow y : \tau \text{ is admitted as a certified trace in V2}\}.$$

*Equivalently, $\mathcal{T}(x, y)$ is the projection of the certified hom-class onto its trace component.*

**Axiom 6.2** (Trace closure)**.** *If $\tau_1 \in \mathcal{T}(x, y)$ and $\tau_2 \in \mathcal{T}(y, z)$, then*

$$\tau_2 \odot \tau_1 \in \mathcal{T}(x, z).$$

**Axiom 6.3** (Identity tightness)**.** *For every boundary state $x$, there exists an identity trace $\mathrm{id}_x \in \mathcal{T}(x, x)$ satisfying*

$$\delta(\mathrm{id}_x) = 0.$$

*Remark* 6.4. Identity tightness is not cosmetic. Without it, one only obtains $d(x, x) \geq 0$, not $d(x, x) = 0$.

## 7. Coh V3 distance

**Definition 7.1** (Coh V3 distance)**.** *For states $x, y$, define*

$$d_{\mathrm{Coh}}(x, y) := \begin{cases} \inf\{\delta(\tau) : \tau \in \mathcal{T}(x, y)\}, & \mathcal{T}(x, y) \neq \emptyset, \\ +\infty, & \mathcal{T}(x, y) = \emptyset. \end{cases}$$

*Remark* 7.2 (Directed extended pseudo-quasi-metric). The word "metric" is used in the Lawvere sense. In general,

$$d_{\mathrm{Coh}}(x, y) \neq d_{\mathrm{Coh}}(y, x),$$

and $d_{\mathrm{Coh}}(x, y) = 0$ need not imply $x = y$. The precise structure is therefore a directed extended pseudo-quasi-metric.

**Lemma 7.3** (Nonnegativity)**.** *For all states $x, y$,*

$$d_{\mathrm{Coh}}(x, y) \geq 0.$$

*Proof.* Each $\delta(\tau)$ is a supremum of nonnegative hidden costs, hence nonnegative. The infimum of a subset of $[0, \infty]$ is nonnegative, and the empty case is defined as $+\infty$. □

**Lemma 7.4** (Identity distance)**.** *For every state $x$,*

$$d_{\mathrm{Coh}}(x, x) = 0.$$

*Proof.* By identity tightness, there exists $\mathrm{id}_x \in \mathcal{T}(x, x)$ with $\delta(\mathrm{id}_x) = 0$. Therefore $d_{\mathrm{Coh}}(x, x) \leq 0$. By nonnegativity, $d_{\mathrm{Coh}}(x, x) \geq 0$. Hence equality holds. □

**Lemma 7.5** (Envelope dominates induced distance)**.** *If $\tau \in \mathcal{T}(x, y)$, then*

$$d_{\mathrm{Coh}}(x, y) \leq \delta(\tau).$$

*Proof.* The value $d_{\mathrm{Coh}}(x, y)$ is the infimum over all values $\delta(\sigma)$ with $\sigma \in \mathcal{T}(x, y)$. Since $\tau$ is one such trace, the infimum is no larger than $\delta(\tau)$. □

**Theorem 7.6** (Directed triangle inequality)**.** *For all states $x, y, z$,*

$$d_{\mathrm{Coh}}(x, z) \leq d_{\mathrm{Coh}}(x, y) + d_{\mathrm{Coh}}(y, z).$$

*Proof.* If either $d_{\mathrm{Coh}}(x, y) = +\infty$ or $d_{\mathrm{Coh}}(y, z) = +\infty$, the right-hand side is $+\infty$, and the claim is immediate.

Assume both are finite. Let $\varepsilon > 0$. By the definition of infimum, choose traces

$$\tau_1 \in \mathcal{T}(x, y), \qquad \tau_2 \in \mathcal{T}(y, z)$$

such that

$$\delta(\tau_1) \leq d_{\mathrm{Coh}}(x, y) + \varepsilon/2, \qquad \delta(\tau_2) \leq d_{\mathrm{Coh}}(y, z) + \varepsilon/2.$$

By trace closure, $\tau_2 \odot \tau_1 \in \mathcal{T}(x, z)$. Hence

$$d_{\mathrm{Coh}}(x, z) \leq \delta(\tau_2 \odot \tau_1).$$

The oplax defect law gives

$$\delta(\tau_2 \odot \tau_1) \leq \delta(\tau_1) + \delta(\tau_2) \leq d_{\mathrm{Coh}}(x, y) + d_{\mathrm{Coh}}(y, z) + \varepsilon.$$

Therefore

$$d_{\mathrm{Coh}}(x, z) \leq d_{\mathrm{Coh}}(x, y) + d_{\mathrm{Coh}}(y, z) + \varepsilon.$$

Letting $\varepsilon \to 0$ proves the claim. □

## 8. Lawvere enrichment

Let $\mathsf{Law}$ denote the monoidal poset

$$([0, \infty], \geq, +, 0).$$

The reversed order is standard in Lawvere metric geometry: the triangle inequality

$$d(x, z) \leq d(x, y) + d(y, z)$$

corresponds to enriched composition in the order $\geq$.

**Theorem 8.1** (Coh V3 enrichment)**.** *Let $\mathcal{C}_{V2}$ be a certified category with trace closure, identity tightness, and the oplax defect law. Then the state-indexed Coh V3 structure is enriched over*

$$([0, \infty], \geq, +, 0).$$

*Equivalently, $d_{\mathrm{Coh}}$ is a Lawvere quasi-metric induced by the V2 certified oplax category.*

*Proof.* The identity axiom for enrichment is $0 \geq d(x, x)$ in the reversed order, equivalent to $d(x, x) \leq 0$, which holds because $d(x, x) = 0$. The composition axiom is

$$d(x, y) + d(y, z) \geq d(x, z),$$

which is exactly the directed triangle inequality. Thus the state-indexed structure is enriched over $([0, \infty], \geq, +, 0)$. □

**Corollary 8.2** (Correct classification)**.** *Coh V3 is not a symmetric metric category and not a weak category. It is a directed, oplax, Lawvere quasi-metric category.*

*Proof.* Asymmetry is permitted because $\mathcal{T}(x, y)$ and $\mathcal{T}(y, x)$ need not be related. Weak categorical associativity is unnecessary because trace composition remains strictly associative at the V2 categorical layer. The only non-strictness is cost-level oplaxness. □

## 9. Geodesics and optimal admissible computation

**Definition 9.1** (Coh geodesic)**.** *A trace $\gamma \in \mathcal{T}(x, y)$ is a Coh geodesic from $x$ to $y$ if*

$$\delta(\gamma) = d_{\mathrm{Coh}}(x, y).$$

*Remark* 9.2. A geodesic is an optimal admissible computation: it realizes the minimal projection-induced defect among certified traces from $x$ to $y$.

**Definition 9.3** (Approximate geodesic)**.** *For $\varepsilon > 0$, a trace $\gamma_\varepsilon \in \mathcal{T}(x, y)$ is an $\varepsilon$-geodesic if*

$$\delta(\gamma_\varepsilon) \leq d_{\mathrm{Coh}}(x, y) + \varepsilon.$$

**Proposition 9.4** (Existence of approximate geodesics)**.** *If $d_{\mathrm{Coh}}(x, y) < \infty$, then for every $\varepsilon > 0$ there exists an $\varepsilon$-geodesic from $x$ to $y$.*

*Proof.* This is the defining property of an infimum over a nonempty set. Since $d_{\mathrm{Coh}}(x, y) < \infty$, the trace set $\mathcal{T}(x, y)$ is nonempty and the infimum is finite. Therefore some trace has defect within $\varepsilon$ of the infimum. □

**Proposition 9.5** (Finite attainment criterion)**.** *Suppose $\mathcal{T}(x, y)$ is finite and nonempty. Then a Coh geodesic from $x$ to $y$ exists.*

*Proof.* The finite set $\{\delta(\tau) : \tau \in \mathcal{T}(x, y)\}$ has a minimum. Any trace attaining this minimum is a Coh geodesic. □

## 10. Metric balls and admissible growth

**Definition 10.1** (Metric ball)**.** *For $r \in [0, \infty]$, define the directed Coh ball*

$$B_r(x) := \{y : d_{\mathrm{Coh}}(x, y) \leq r\}.$$

**Definition 10.2** (Growth profile)**.** *If the observable state set is countable or discretized at fixed observational resolution, define the growth profile*

$$N_x(r) := |B_r(x)|.$$

**Definition 10.3** (Upper Coh growth dimension)**.** *When $N_x(r)$ is finite for sufficiently large finite $r$, define*

$$D^+_{\mathrm{Coh}}(x) := \limsup_{r \to \infty} \frac{\log N_x(r)}{\log r}.$$

*Remark* 10.4. Growth dimension is not asserted to be finite in general. It depends on the state representation, verifier acceptance, projection resolution, and the hidden cost envelope.

## 11. Failure modes and boundary cases

11.1. **Infinite fibers.** If

$$\sup_{\xi \in \mathcal{F}_R} W(\xi) = +\infty,$$

then $\delta(R) = +\infty$. Any distance whose admissible paths all require such traces becomes infinite. This is not a contradiction; it records non-finite certification.

11.2. **Degenerate verifier.** If the verifier admits all proposed traces and the envelope can be made arbitrarily small between many states, then many distances collapse toward zero. The result is a pseudo-quasi-metric with weak separation.

11.3. **Projection collapse.** If $\Pi$ forgets too much information, the fiber envelope may become too loose to provide useful geometry. The formal theorem remains true, but the induced geometry may be uninformative.

11.4. **No identities with zero defect.** If identity tightness fails, $d(x, x) = 0$ is not guaranteed. This is why identity tightness is part of the V3 specification.

11.5. **Boundary case: strict additivity.** If

$$\delta(\tau_2 \odot \tau_1) = \delta(\tau_1) + \delta(\tau_2)$$

for all composable traces, then the system behaves like a strict additive cost category at the envelope level. Coh allows but does not require this. Projection usually makes the inequality strict.

## 12. Lean and implementation alignment

A Lean formalization of V3 should separate the following layers.

(1) A type of states.
(2) A type of admissible traces indexed by endpoints.
(3) A defect function $\delta$ taking traces to an extended nonnegative domain.
(4) A composition operation preserving endpoint typing.
(5) Axioms or theorems for identity tightness and defect subadditivity.
(6) A distance definition as an infimum over endpoint-indexed trace sets.
(7) Proofs of identity and triangle inequality.

In executable verifiers, real suprema are replaced by conservative computable envelopes. The proof obligation is not that the implementation computes the exact supremum, but that the declared envelope dominates all hidden realizations allowed by the projection model.

## 13. Conclusion

Coh V3 is the geometry layer of the Coh stack. V1 supplies verifier admissibility. V2 supplies a projection-certified category and an oplax envelope defect. V3 defines distance by minimizing that envelope over admissible traces. Under trace closure, identity tightness, and the oplax defect law, this distance satisfies identity and the directed triangle inequality and therefore yields a Lawvere quasi-metric enrichment over $([0, \infty], \geq, +, 0)$.

The central contribution is conceptual and structural:

$$\boxed{\text{Geometry is induced by verifier-constrained worst-case hidden cost.}}$$

This makes V3 a geometry of certified computation rather than a metric imposed from outside.

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| Result | Depends on | Status |
|---|---|---|
| Envelope soundness | definition of $\delta$ | proved |
| Oplax defect | fiber decomposition, hidden subadditivity | proved |
| V2 category | oplax defect, resource inequality, trace closure | inherited |
| Identity distance | identity tightness | proved |
| Triangle inequality | trace closure, oplax defect, infimum | proved |
| Lawvere enrichment | identity and triangle inequality | proved |
| Geodesic existence | finite or compact trace domain | conditional |

TABLE 2. Logical dependency map for the V3 paper.

## APPENDIX A. DEPENDENCY MAP

## APPENDIX B. MINIMAL EXAMPLE

Let $X = \{A, B, C\}$ and suppose the admissible trace sets contain

$$\tau_{AB} : A \rightsquigarrow B, \qquad \tau_{BC} : B \rightsquigarrow C,$$

with

$$\delta(\tau_{AB}) = 1, \qquad \delta(\tau_{BC}) = 2.$$

If the composite is admissible and satisfies

$$\delta(\tau_{BC} \odot \tau_{AB}) \leq 3,$$

then

$$d(A, C) \leq 3.$$

If no admissible trace exists from $C$ to $A$, then

$$d(C, A) = +\infty.$$

Thus the induced geometry is directed.